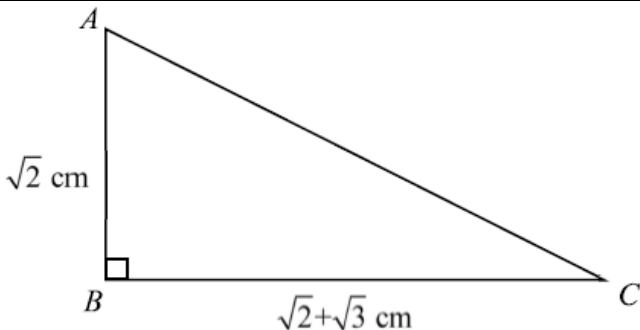


1	ACSBR/2025/II/1
	Given that $\frac{\sqrt{a}+\sqrt{b}}{9}$ is the root of the equation $x\sqrt{30} = x\sqrt{3} + \sqrt{6}$, find the value of a and of b, where a and b are integers and $a > b$. [3]
2	CCHY/2025/I/1
	 The diagram above shows a right-angled triangle ABC, where $AB = \sqrt{2}$ cm, and $BC = \sqrt{2} + \sqrt{3}$ cm. (a) Find the exact value of $\tan \angle ACB$ in the form of $a + \sqrt{b}$, where a and b are integers. [2] (b) Hence, find the exact value of $\sec^2 \angle ACB$ in the form $m + n\sqrt{k}$, where m and n are integers. [2]
3	GMS/2025/I/7
	(a) A parallelogram with base $(4 - \sqrt{12})$ m has an area of $(22 - \sqrt{48})$ m². Without using a calculator, find the height, in m, of the parallelogram in the form $(p + q\sqrt{3})$, where p and q are integers. [3] (b) The length, in m, of each side of a square with an area of $(49 + 20\sqrt{6})$ m² can be expressed in the form $(\sqrt{c} + \sqrt{d})$ where c and d are integers and $c < d$. Find the value of c and of d. [3]
4	ZHSS/2025/I/2
	The base of a cuboid has an area of $(1 - \sqrt{2})^2$ cm². The volume of the cuboid is $(16 + 3\sqrt{2})$ cm³. Without using a calculator, express the height of the cuboid in the form $a + b\sqrt{2}$, where a and b are constants. [5]

1	$x = \frac{\sqrt{6}}{\sqrt{30}-\sqrt{3}} \times \frac{\sqrt{30}+\sqrt{3}}{\sqrt{30}+\sqrt{3}}$ $= \frac{\sqrt{180}+\sqrt{18}}{27}$ $= \frac{3\sqrt{20}+3\sqrt{2}}{27}$ $= \frac{\sqrt{20}+\sqrt{2}}{9}$ $a=20, b=2$
2	(a) $\tan \angle ACB = \frac{\sqrt{2}}{\sqrt{2}+\sqrt{3}} \times \frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}-\sqrt{3}}$ $= \frac{2-\sqrt{6}}{2-3}$ $= -2+\sqrt{6}$ (b) $\sec^2 A = 1 + \tan^2 A$ $= 1 + (\sqrt{6}-2)^2$ $= 1 + 6 - 4\sqrt{6} + 4$ $= 11 - 4\sqrt{6}$
3	(a) $\text{Height} = \frac{22-4\sqrt{3}}{4-2\sqrt{3}} \times \frac{4+2\sqrt{3}}{4+2\sqrt{3}}$ $= \frac{(22-4\sqrt{3})(4+2\sqrt{3})}{4^2-4(3)}$ $= \frac{1}{4}(88+44\sqrt{3}-16\sqrt{3}-24)$ $= \frac{1}{4}(64+28\sqrt{3})$ $= 16+7\sqrt{3}$ (b)

	$(\sqrt{c} + \sqrt{d})^2 = 49 + 20\sqrt{6}$ $c + d + 2\sqrt{cd} = 49 + 20\sqrt{6}$ $c + d = 49$ $d = 49 - c \text{ -----(1)}$ $2\sqrt{cd} = 20\sqrt{6}$ $cd = 600 \text{ -----(2)}$ Compare rational/irrational terms $c(49 - c) = 600$ $c^2 - 49c + 600 = 0$ $(c - 25)(c - 24) = 0$ $c < d, \quad c = 24, \quad d = 25$
4	$\text{height} = \frac{16 + 3\sqrt{2}}{(1 - \sqrt{2})^2}$ $= \frac{16 + 3\sqrt{2}}{3 - 2\sqrt{2}} \times \frac{3 + 2\sqrt{2}}{3 + 2\sqrt{2}}$ $= \frac{(16 + 3\sqrt{2})(3 + 2\sqrt{2})}{9 - 8}$ $= 48 + 9\sqrt{2} + 32\sqrt{2} + 12$ $= 60 + 41\sqrt{2}$